

in the *Version* drop-down menu. This selection helps in the auto selection of the type of storage interface in the next *Hard Disk* tab. If Microsoft Windows is selected, the default is the IDE interface and, if it is Linux, the default is VirtIO SCSI.

- In the *Storage* tab, select *zfsdata* as the storage location for both Windows10VM and Ubuntu18VM. Also allocate 100GB as *Disk Size*.
- In the next *CPU* tab, allocate two cores for the VM and in the *Memory* tab, allocate 4096, i.e., 4GB RAM.
- In the *Network* tab, let the defaults settings remain. Go to the *Confirm* tab, which shows a summary of what has been selected and click on the *Finish* button. Now in a few minutes, the VM gets created.
- Start the VM and access its console. Click on the *Start* button to start the VM and to access the GUI, click on the *Console* button. Here you can finish the OS installation as per standard MS Windows and Ubuntu Linux installation processes, including assigning IP, etc, as if it's a part of the LAN setup.
- Once the VM is up, you can use it over your LAN via a remote desktop or the Proxmox Web-console interface.
- For Windows to perform better, one can add additional VM drivers and follow the best practices guidelines given by the Proxmox Wiki (https://pve.proxmox.com/wiki/Windows_10_guest_best_practices).

Steps to schedule replication

To replicate the VMs created, we need to configure the replication scheduler to replicate from the pve01 to the pve02 server.

- Click on the *Replication* section shown in each VM's sub-menu. For example, click on the Ubuntu18VM VM from the left-menu and on the right-side panel you will see a sub-menu, which has an option called *Replication*, which you can click on.
- Then click the *Add* button at the top; a pop-up with the words *Create:Replication job with target as pve02* will appear, as the existing VM is on pve01.
- Now select *Every 30 min* in the *Schedule* section and click on the *Create* button.

After creation, an entry is seen in the replication job list. Now select the job and click on *Schedule*, which will start taking a time copy to pve02, and the technical details will be seen in the *Log* section.

Migration from pve01 to pve02 and vice versa

After the replication setup is done, we can migrate the VMs from one server to another as follows:

- Click the *Migrate* button shown at the top section of the VM, next to the *Console* button, to migrate the VM from pve01 to pve02. The moment the VM is migrated to pve02, the replication target is also changed from pve02 to pve01. So, after a few days, we can move back to pve01.

- An IT admin can keep VMs running on pve01 for the first and third weeks of the month, and on pve02 for the second and fourth weeks of the month. Thus both pieces of hardware are used and VMs are switched between the servers.

Backup of VMs to external Windows Share drives

We now need to add Windows Share as the backup storage in the *Storage* section under *Data centre*.

- Click on the *Add* button under *Storage*, and you will be asked for many options. Select CIFS for using the Windows Share drive.
- A pop-up called *Add:CIFS* will open, asking you to enter a name in the *ID* section. Name it as *winbackupdrive*, and in *Server*, enter the IP address. In our case it is 192.168.1.2. Next, enter the user name and password.
- After that, it will populate Windows shares. Under the 'Share' dropdown, select *Share "vmbbackup"* for VM backup storage.
- Then select the *VZDump* backup file in the *Content* drop-down and also change *Max Backups* to 5, as we need to maintain five full copies of VMs for safety purposes.

Now we select *Backup from Data centre* in the right-side section, and click on *Add* to add a schedule to take backups of both VMs.

ID	Node	Status	Name	Type
100	pve01	stopped	WINDOWS10VM	qemu
101	pve01	stopped	UBUNTU18VM	qemu

Figure 3: Adding backup schedule

- In the pop-up window, select *winbackupdrive* in the *Storage* section with *Days of week* as *Saturday* and *Start Time* as 19:00 and, finally, select both VMs from the list and click on the *Create* button.

So now we can create a full VM backup every Saturday, which can be restored in the event of a crash as a different or the same VM on this cluster or on another Proxmox cluster. **END** 🐧

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